Day 78 - Azure Key Vault

Objective

To understand and implement Azure Key Vault for securely storing secrets and keys, and to use them in an application.

Steps Performed

Created Azure Key Vault (KushVault) in Azure Portal.

Configured access policies and permissions.

Created a Secret inside Key Vault.

Created a Key inside Key Vault.

Accessed Key Vault using .NET Console Application.

Used DefaultAzureCredential for authentication.

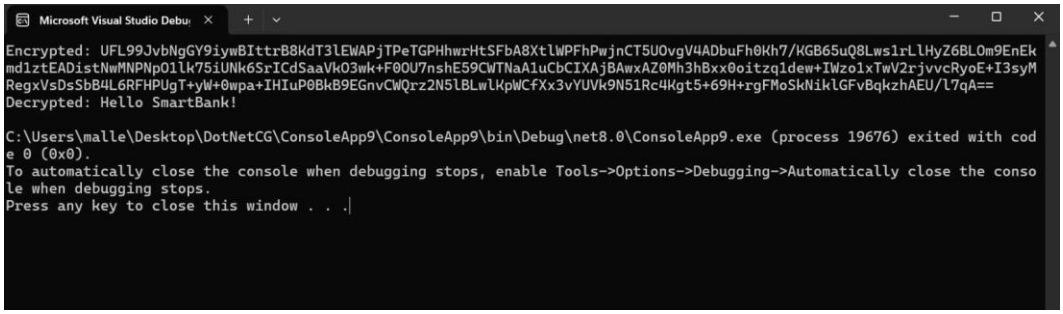
Implemented encryption and decryption using Azure Key Vault keys.

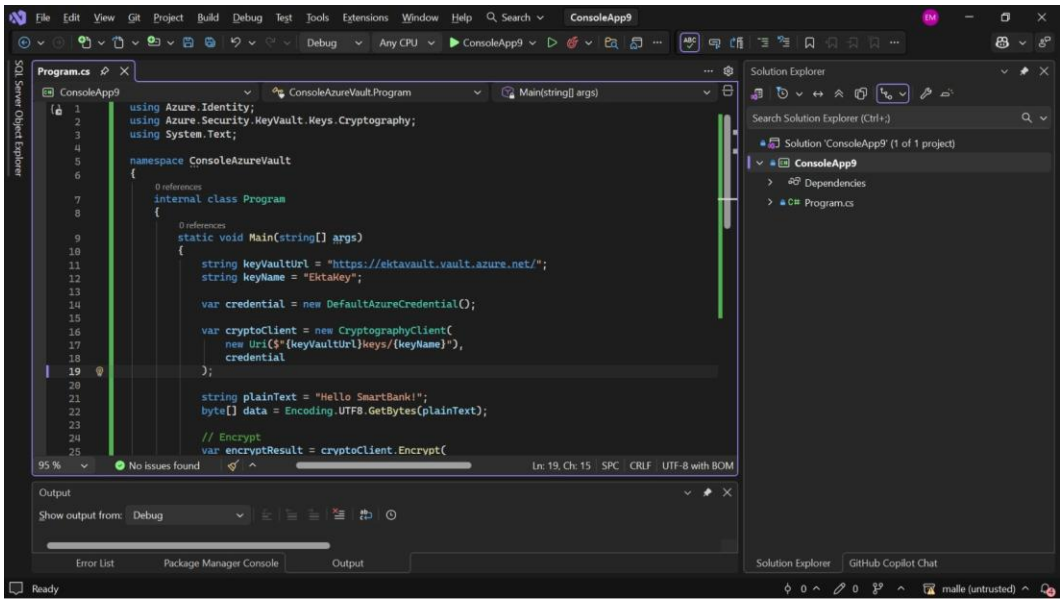
Verified output in console.

Implementation (Code Summary)

Used Azure.Security.KeyVault.Keys.Cryptography with DefaultAzureCredential to perform encryption and decryption.

Screenshots




File Edit View Git Project Build Debug Test Tools Extensions Window Help Search ConsoleApp9
Debug Any CPU ConsoleApp9
Program.cs
ConsoleApp9 ConsoleAzureVault.Program Main(string[] args)
using Azure.Identity;
using Azure.Security.KeyVault.Keys.Cryptography;
using System.Text;
namespace ConsoleAzureVault
{
0 references
internal class Program
{
0 references
static void Main(string[] args)
{
string keyVaultUrl = "https://ektavault.vault.azure.net/";
string keyName = "EktaKey";
var credential = new DefaultAzureCredential();
var cryptoClient = new CryptographyClient(
new Uri($"{keyVaultUrl}keys/{keyName}"),
credential
);
string plainText = "Hello SmartBank!";
byte[] data = Encoding.UTF8.GetBytes(plainText);
// Encrypt
var encryptResult = cryptoClient.Encrypt(
95 % No issues found Ln: 19, Ch: 15 SPC CRLF UTF-8 with BOM
Output
Show output from: Debug
Error List Package Manager Console Output
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'ConsoleApp9' (1 of 1 project)
ConsoleApp9
Dependencies
Program.cs
Solution Explorer GitHub Copilot Chat
Ready
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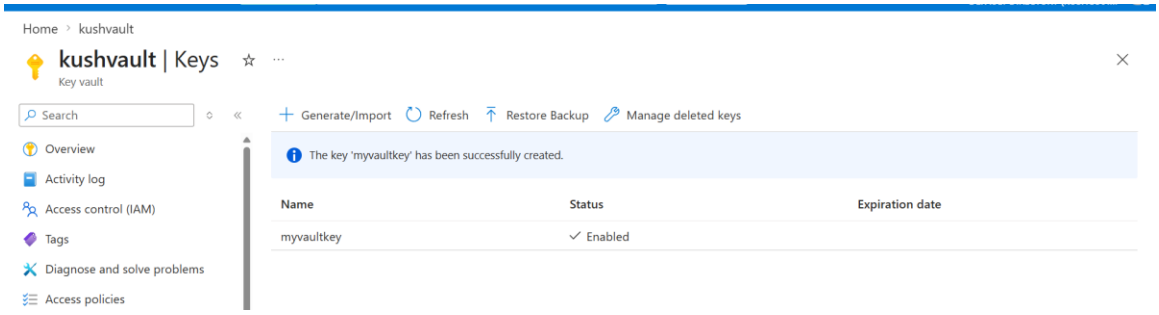
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Overview
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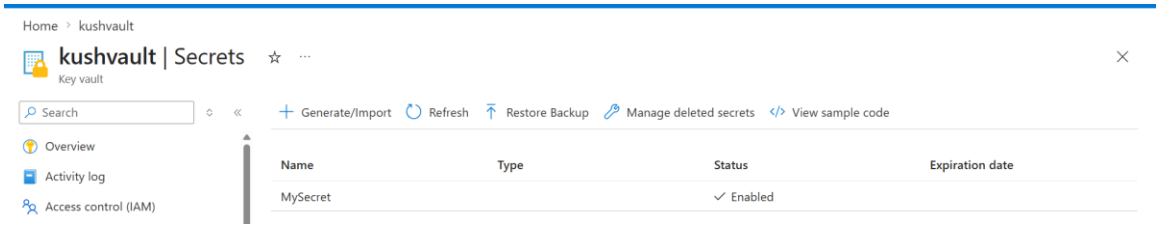
Diagnose and solve problems
Access policies
Generate/Import Refresh Restore Backup Manage deleted keys
The key 'myvaultkey' has been successfully created.
Name Status Expiration date
myvaultkey Enabled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MySecret Enabled